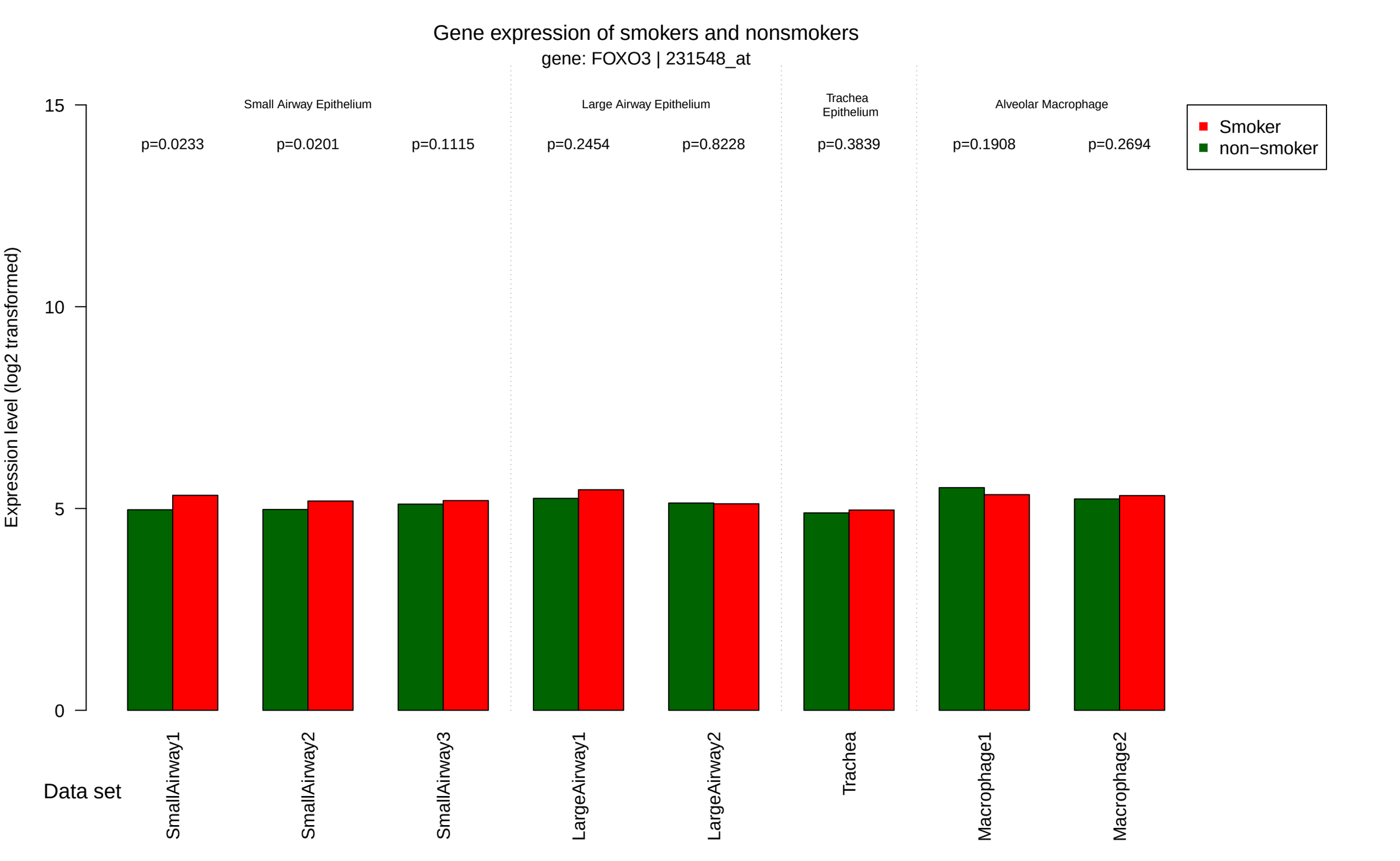
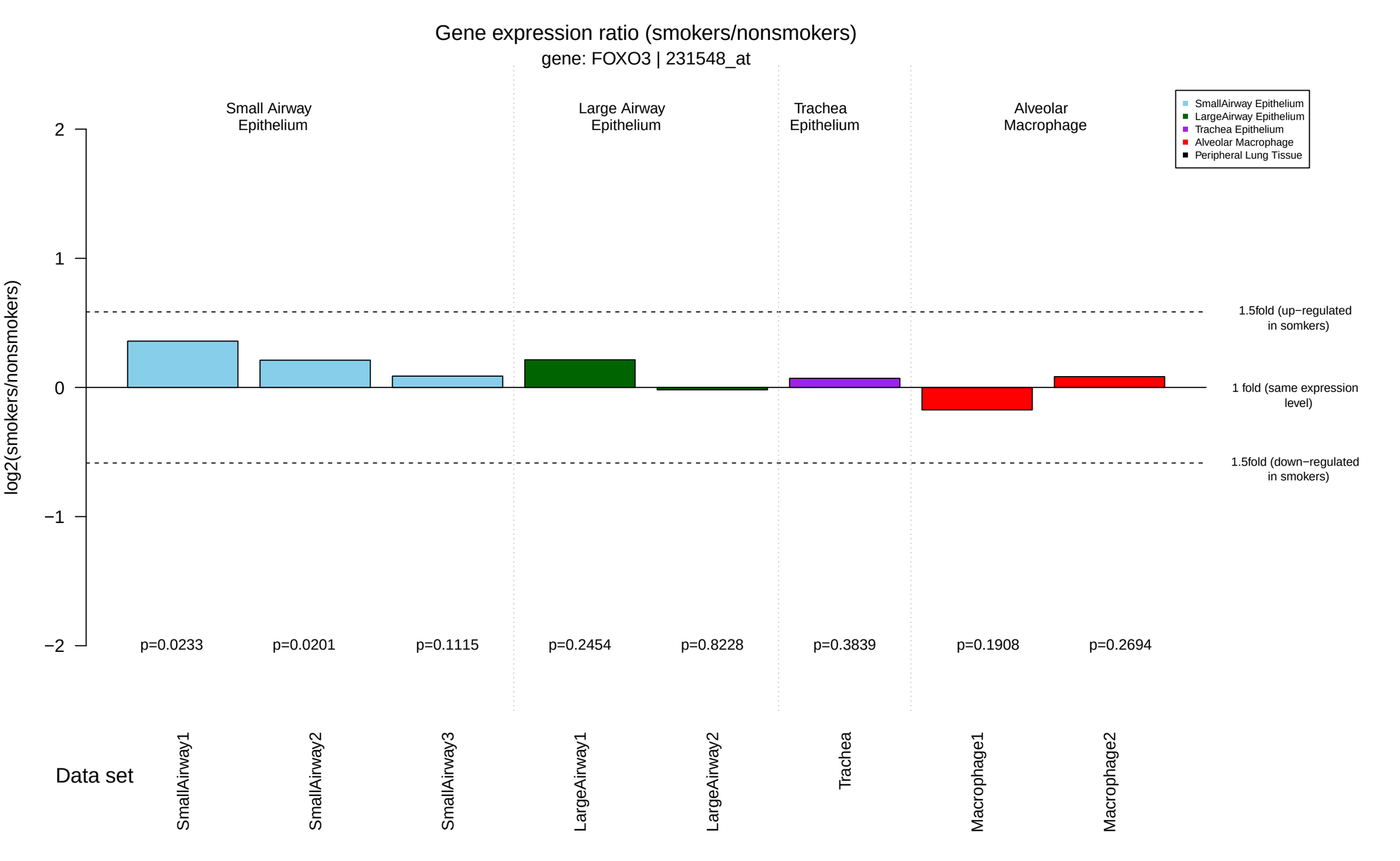
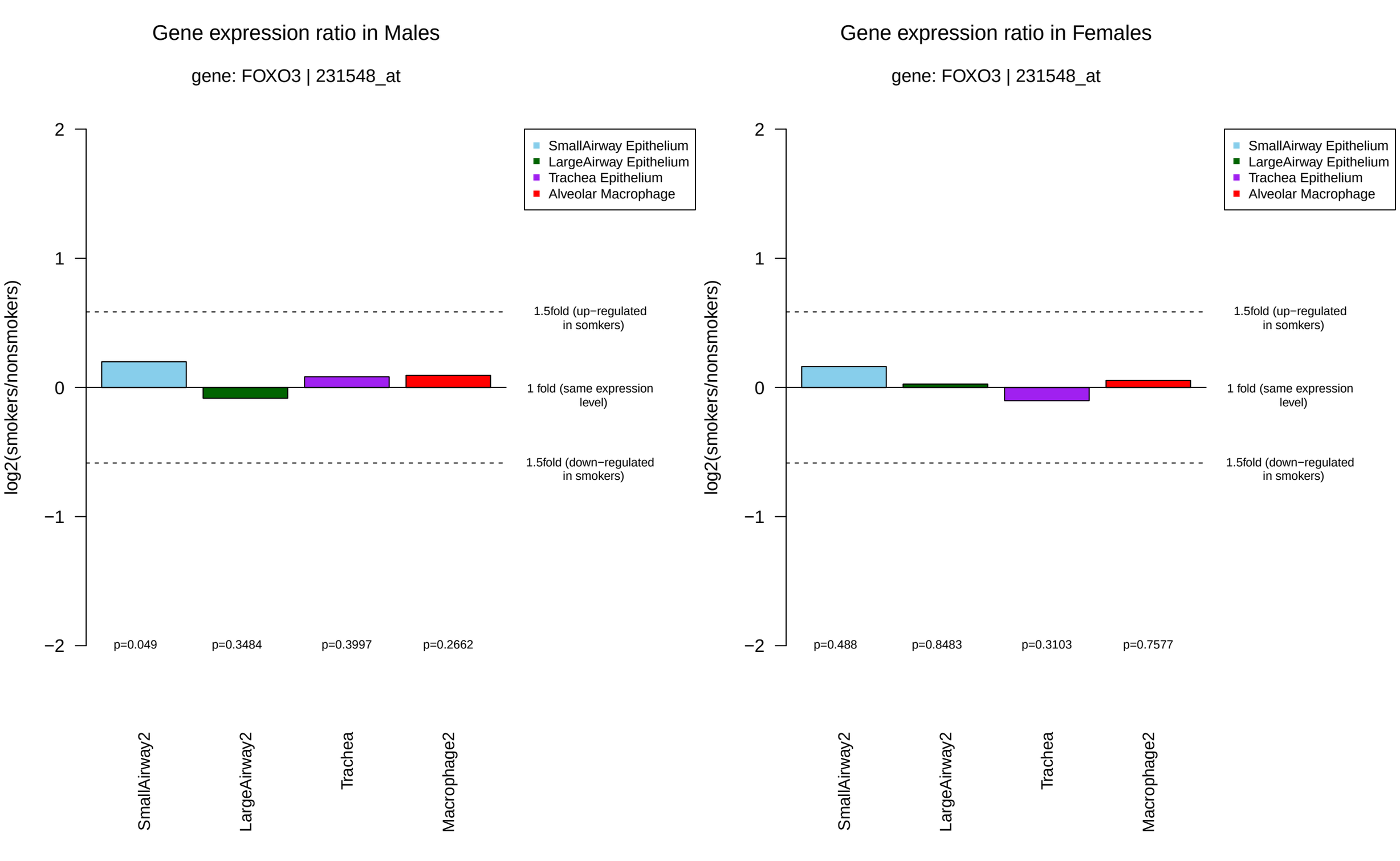
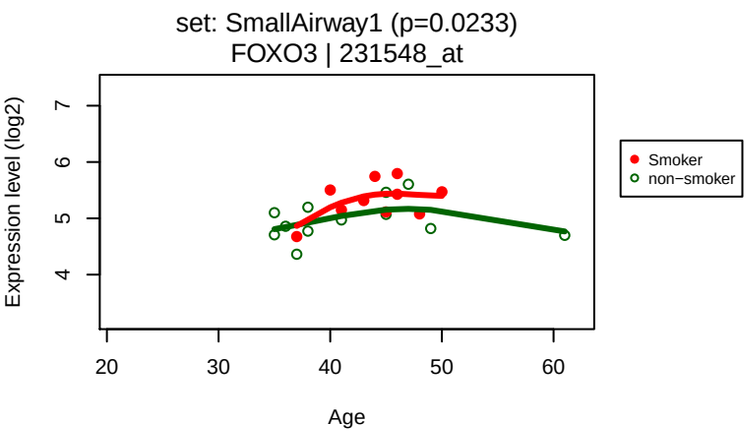
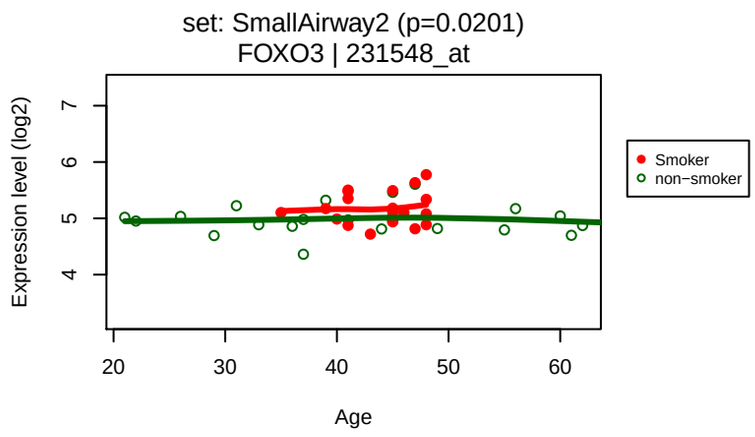
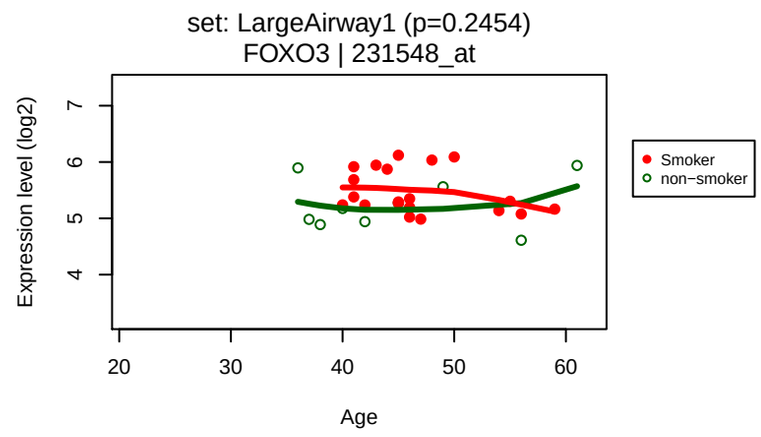

Gene expression of smokers and nonsmokers
gene: FOXO3 | 231548_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0233
p=0.0201
p=0.1115
p=0.2454
p=0.8228
p=0.3839
p=0.1908
p=0.2694
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

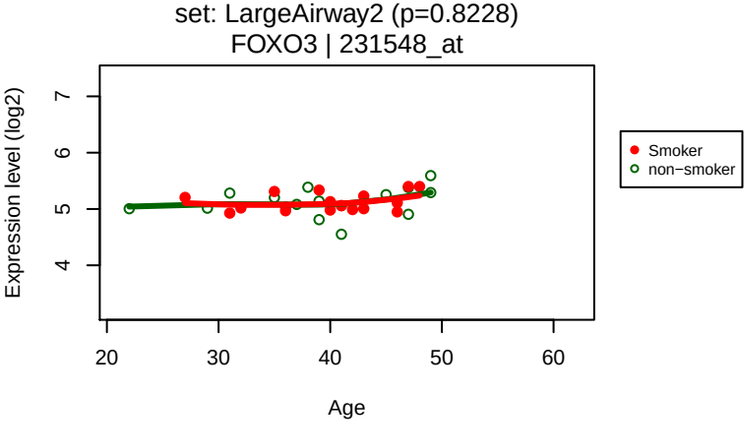

Gene expression ratio (smokers/nonsmokers)
gene: FOXO3 | 231548_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0233
p=0.0201
p=0.1115
p=0.2454
p=0.8228
p=0.3839
p=0.1908
p=0.2694
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

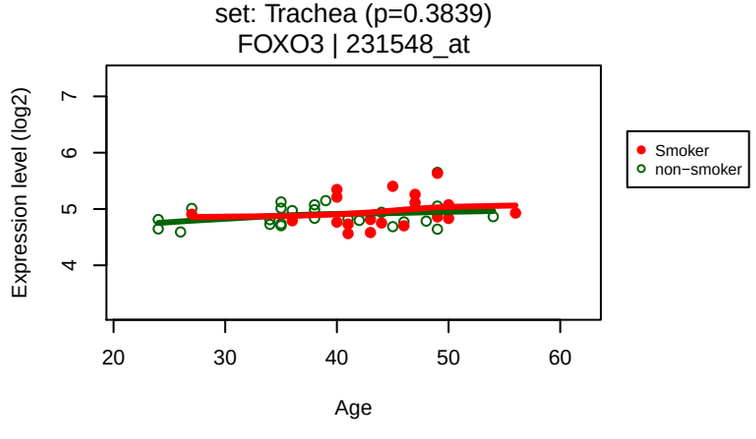

Gene expression ratio in Males
gene: FOXO3 | 231548_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.049
p=0.3484
p=0.3997
p=0.2662
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: FOXO3 | 231548_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.488
p=0.8483
p=0.3103
p=0.7577
SmallAirway2
LargeAirway2
Trachea
Macrophage2

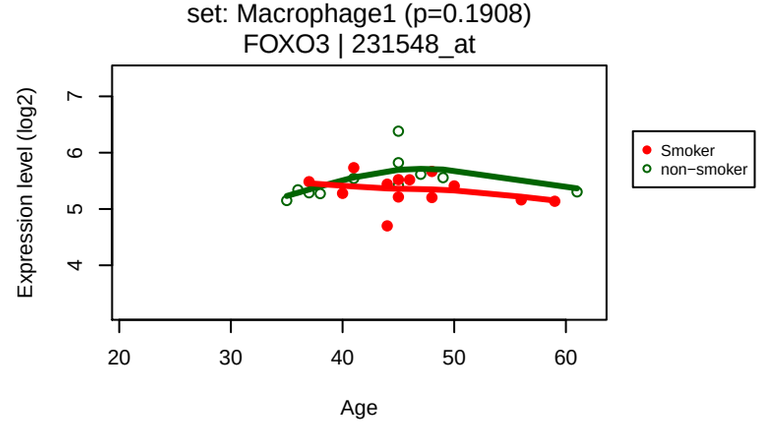

set: SmallAirway1 (p=0.0233)
FOXO3 | 231548_at
Expression level (log2)
Age
Smoker
non−smoker

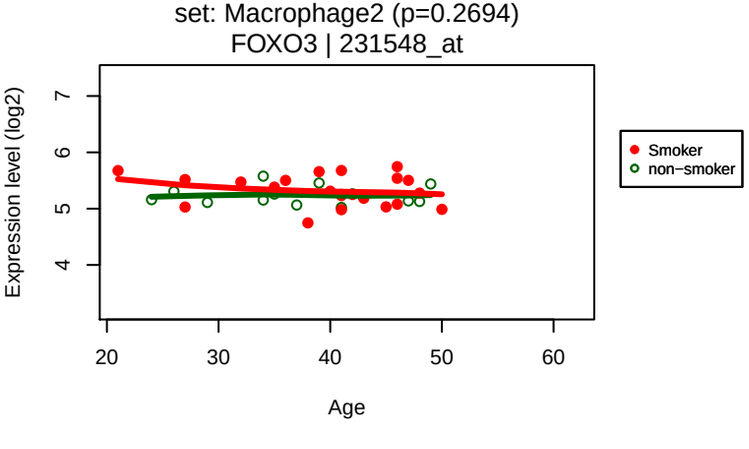

set: SmallAirway2 (p=0.0201)
FOXO3 | 231548_at
Expression level (log2)
Age
Smoker
non−smoker

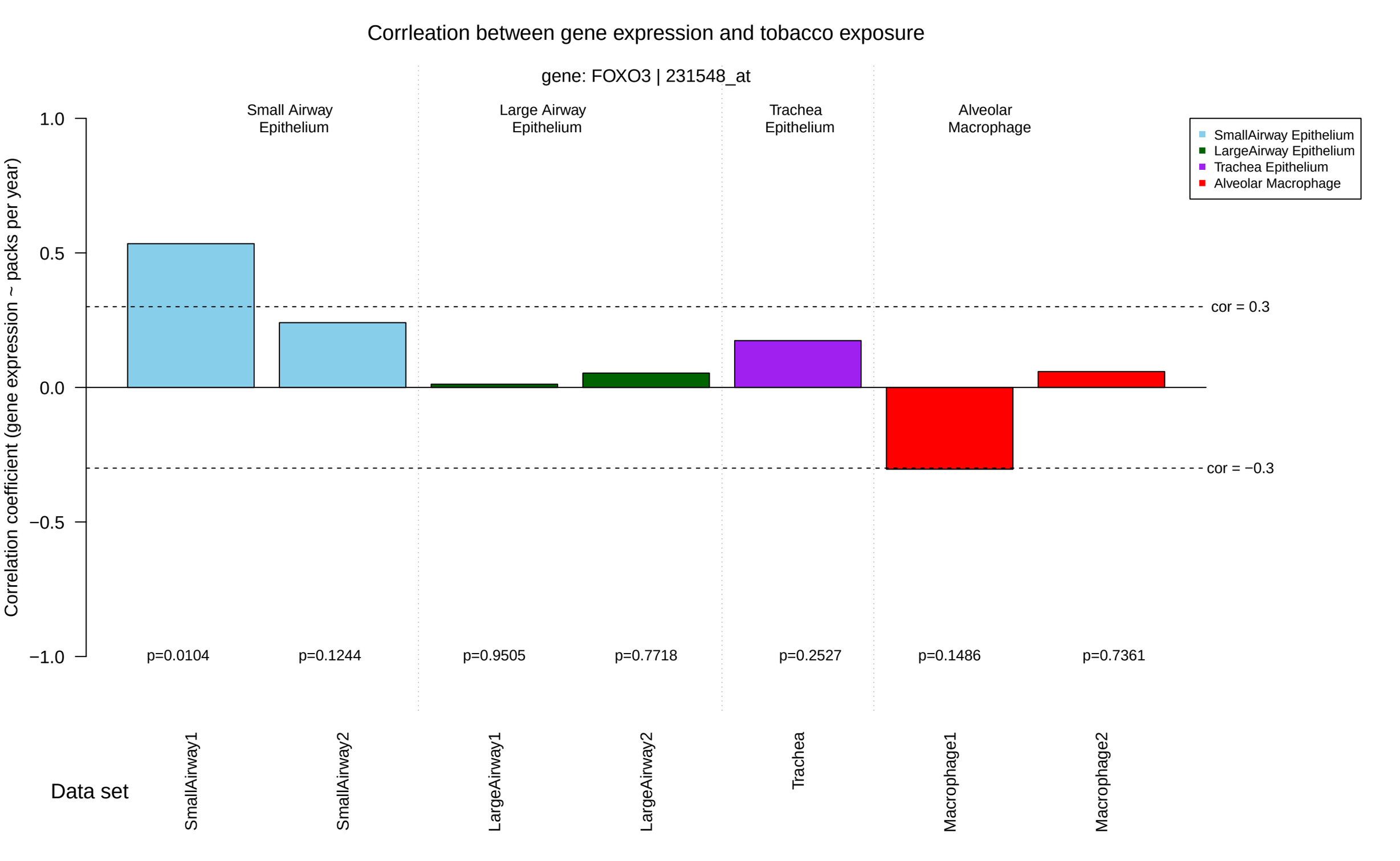

set: LargeAirway1 (p=0.2454)
FOXO3 | 231548_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8228)
FOXO3 | 231548_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3839)
FOXO3 | 231548_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1908)
FOXO3 | 231548_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2694)
FOXO3 | 231548_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: FOXO3 | 231548_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0104
p=0.1244
p=0.9505
p=0.7718
p=0.2527
p=0.1486
p=0.7361
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2